

Alikasim Budhwani

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EXPERIENCE

• BHF Robotics [🌐]

Robotics Engineer

Sep 2023 - Dec 2024

Markham, Canada

- Led team of 4 in developing ROS-based autonomy stack for Agrobot mobile platform which conducts targeted weed elimination via high-voltage electricity.
- Implemented ant colony optimization (ACO) for weed elimination sequencing, achieving 300 weeds/min cycle time.
- Developed ROS perception module for precise weed detection and elimination site selection, minimizing crop damage while achieving 90% accuracy.

• Ansys [🌐]

SDE Intern

Jan 2023 - Apr 2023

Waterloo, Canada

- Implemented isoparametric mapping in C++-based finite element solver, reducing computation time by 10% for 10,000+ node models with complex curved geometries.
- Refactored C++ element classes in coupling framework using templates, eliminating 15+ redundant classes and improving maintainability.
- Integrated unit tests (C++) in CI/CD pipeline to verify the functionality, and edge cases of higher-order finite mesh elements up to order 4.

• Kinectrics [🌐]

R&D Intern

May 2022 - Aug 2022

Etobicoke, Canada

- Developed end-to-end robotic solutions for waste decommissioning of irradiated pressure vessels at Pickering Nuclear Station.
- Researched, and validated UAV solutions for radiation monitoring and line inspection pilot projects.

• Promotion [🌐]

Perception Intern

Sep 2021 - Dec 2021

Oakville, Canada

- Developed an in-situ defect detection system for 3D metal printing, obtaining 75% detection accuracy.
- Implemented a custom hand-eye calibration algorithm for 2D laser scanners, reaching 0.5mm accuracy during 3D object reconstruction.
- Designed a vision-based trajectory adjustment algorithm for automotive robots, achieving 0.2mm recalibration error.
- Created a semi-supervised data processing pipeline to segment printed samples, filter noise, and annotate point clouds, reducing time by 60%.

EDUCATION

• University of Waterloo

MASc in Mechanical Engineering (GPA: 4.00/4.00)

Jan 2025 - Apr 2027

Waterloo, Canada

- **Research:** Leveraging chaotic dynamics to learn safety filters for contact-rich manipulation policies.
- **Coursework** including Nonlinear Control, Stochastic Control, and Vehicle Dynamics.
- Recipient of Ontario Graduate Scholarship (OGS).

• University of Waterloo

BASc in Mechanical Engineering (GPA: 3.84/4.00)

Sep 2019 - Aug 2024

Waterloo, Canada

- **Coursework** including Robot Manipulation, Mobile Robotics, Reinforcement Learning and Multivariable Control.

SKILLS

- **Programming Languages:** Python, C++, MATLAB, R, Bash
- **Toolboxes/Environments:** ROS, Linux, Docker, PyTorch, Git, MuJoCo